

MAGIC VALLEY, ID, USA

WMO: 725866

Lat: 42.482N

Lon: 114.487W

Elev: 4151

StdP: 12.62

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 94178

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.0	11.6	-0.5	6.2	10.3	4.1	7.9	15.7	33.4	36.5	29.4	34.8	6.6	100	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.5	95.0	62.6	92.1	62.0	89.7	61.4	66.1	88.5	64.4	86.1	63.0	84.5	8.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.2	84.4	74.5	55.5	76.5	74.3	54.1	72.7	73.7	33.0	88.7	31.6	85.9	30.4	84.5	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	24.5	20.8	0.1	100.2	6.6	2.2	-4.7	101.8	-8.5	103.1	-12.2	104.4	-17.0	106.0		
(5)			-0.7	69.5	6.5	2.1	-5.3	71.0	-9.1	72.2	-12.7	73.4	-17.4	74.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	30.2	34.0	42.1	47.6	56.4	65.4	75.2	72.9	63.6	50.5	38.5	29.5		
	DBStd	17.37	8.69	7.92	7.29	7.35	8.22	7.67	5.50	5.54	7.74	8.41	8.96	9.20		
	HDD50	2581	613	449	262	132	32	2	0	0	6	95	355	635		
	HDD65	6036	1078	869	710	524	287	89	5	10	117	453	794	1100		
	CDD50	2793	0	1	17	59	230	462	782	709	412	111	10	0		
	CDD65	773	0	0	0	1	21	100	321	253	73	4	0	0		
	CDH74	9746	0	0	1	34	346	1386	3935	2995	974	75	0	0		
	CDH80	4368	0	0	0	4	91	544	1997	1392	325	15	0	0		
	WSAvg	10.4	9.5	11.0	11.7	12.2	10.9	10.6	9.7	9.5	9.6	10.3	9.8	10.0		
	PrecAvg	10.7	1.1	0.7	0.9	1.3	1.5	0.8	0.4	0.5	0.6	0.9	0.9	1.1		
	PrecMax	17.1	2.4	1.7	1.9	3.9	4.1	4.6	0.8	5.3	2.5	2.8	2.0	3.2		
(15) Precipitation	PrecMin	5.7	0.3	0.2	0.3	0.2	0.3	0.0	0.0	0.0	0.0	0.1	0.3	0.0		
	PrecStd	3.2	0.6	0.4	0.5	0.8	1.1	0.9	0.2	1.1	0.6	0.7	0.5	0.8		
	0.4%	DB	53.5	58.5	71.7	78.6	87.8	94.3	99.5	97.2	91.4	81.5	66.5	55.9		
	MCWB	43.9	44.6	49.3	54.0	59.0	62.9	65.6	63.6	61.4	55.4	48.9	44.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	47.9	53.0	64.9	72.6	82.0	90.1	96.4	93.4	87.6	74.9	61.2	50.2		
	MCWB	40.4	42.8	47.3	51.3	57.8	60.9	63.0	62.2	59.9	53.5	46.8	42.6			
	5%	DB	44.9	48.6	60.7	67.4	77.8	86.2	93.3	91.0	83.7	70.2	56.9	45.9		
	MCWB	38.7	40.7	45.3	49.7	55.5	59.7	62.4	61.7	58.3	51.4	44.9	39.5			
	10%	DB	41.5	45.4	55.4	62.7	72.6	82.3	90.6	88.3	80.1	65.6	52.5	42.6		
	MCWB	36.3	39.0	43.3	47.5	54.0	58.4	61.9	61.1	56.9	49.3	42.7	37.2			
	0.4%	WB	45.0	46.3	51.1	56.0	62.4	66.1	69.0	67.7	65.7	57.4	51.2	45.7		
	MCDB	51.2	55.3	65.2	74.4	81.7	90.0	91.8	91.3	82.7	76.3	62.7	52.2			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.5	43.9	48.5	52.7	59.3	63.5	66.6	65.5	62.5	54.6	48.2	42.9		
	MCDB	46.9	51.5	62.3	68.7	77.2	85.3	89.2	87.8	81.4	71.3	59.0	50.0			
	5%	WB	39.1	41.7	46.5	50.4	57.1	61.2	65.0	63.9	60.1	52.6	45.8	40.3		
	MCDB	43.9	48.1	58.7	65.1	74.1	81.9	87.1	85.1	78.9	67.2	55.4	45.5			
	10%	WB	37.0	39.4	44.3	48.2	54.9	59.3	63.5	62.2	58.1	50.3	43.6	37.4		
	MCDB	41.0	45.0	54.4	61.1	70.8	79.2	85.5	83.7	77.2	64.0	51.6	41.8			
	5% DB	MDBR	14.2	15.6	20.5	22.1	23.9	26.4	28.5	28.2	26.8	23.5	19.8	14.8		
	MCDBR	17.3	19.4	28.2	30.7	30.8	32.3	31.6	32.2	32.4	29.8	25.7	17.0			
(36) Mean Daily Temperature Range	MCWBR	11.7	12.4	14.9	15.2	13.9	14.3	13.2	13.9	14.9	15.0	14.5	11.2			
	5% WB	MCDBR	16.0	18.1	25.6	27.5	28.3	31.0	30.0	30.0	28.9	26.5	23.4	16.1		
	MCWBR	11.3	12.0	14.2	14.3	13.5	14.1	13.1	13.6	13.9	13.9	14.0	11.1			
	taub	0.250	0.260	0.273	0.295	0.315	0.311	0.325	0.323	0.297	0.276	0.258	0.250			
(40) Clear-Sky Solar Irradiance	taud	2.573	2.589	2.572	2.501	2.473	2.525	2.495	2.478	2.547	2.590	2.595	2.576			
	Ebn at Noon	293	307	312	310	304	304	298	296	299	295	287	284			
	Edh at Noon	21	24	28	32	34	33	33	33	28	24	20	19			
	RadAvg	568	835	1275	1740	2098	2426	2390	2079	1671	1135	692	452			
(44) All-Sky Solar Radiation	RadStd	68	103	107	119	184	163	131	103	98	75	61	67			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	-1.28	-1.60	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page